Uncertainty Visualisation for Signal Supression

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Abstract

The abstract should outline the main approach and findings of the thesis and must not be more than 500 words.

Declaration

Use only one of the following declarations (Standard thesis or Thesis including published works declaration) and remove the other.

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Publications during enrolment

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The material in has been submitted to the journal *Journal of Impossible Results* for possible publication.

The contribution in __ of this thesis was presented in the International Symposium on Nonsense held in Dublin, Ireland, in July 2022.

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(The inclusion of co-authors reflects the fact that the work came from active collaboration between researchers and acknowledges input into team-based research.)

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Thesis chapter	Publication title	Status	Nature and % of student contribution	Nature and % of coauthors' contribution	Coauthors are Monash students
2	The life cycle of Mongolian crickets	Submitted	Concept and data analysis, writing first draft: 60%	Shu Xu, input into manuscript: 25%; Eddie Betts, input into manuscript: 15%	Shu Xu: No; Eddie Betts: Yes

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Acknowledgements

I would like to thank my pet goldfish for . . .

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Chapter 1

Introduction

- overview of thesis
- what is this thesis about (what problem is it solving)
- how do the chapters fit together
- Uncertainty visualisation is important
  - Decision making
  - correctly hedge conclusions drawn from graphics
- Uncertainty visualisation is a relatively new field, but it has been poorly defined
- Uncertainty itself is a vague term, each paper seems to take the meaning of it as a given, so the visualisation of it is fraught with disagreement
- Two distinct camps have established themselves in the literature. The first view uncertainty visualisation as a visualisation of an error, probability, or distribution. They treat uncertainty as a variable of interest, distinct from its estimate. Examples of this approach include ___.
- The second camp sees uncertainty as noise. They believe uncertainties role in visualisation is to dampen misleading or false conclusions, and amplify correct conclusions. They treat uncertainty as a necessary nuisance that should be communicated alongside it's estimate. Examples of this approach include ___.
- These distinct camps create several gaps in the literature
- Uncertainty visualisation are often bespoke and are not properly integrated into the grammar of graphics. The elusive definition of uncertainty prevents it from being integrated into the grammar of grpahics, which means its role in a visualisation is never clear.

- This leads to visualisation approaches that jump randomly between viewing uncertainty as a variable in or itself, or as a necessary nuisance required to property visualise an estimate. This leads software that visualise uncertainty behaving inconsistently when deciding how to display uncertainty in the graphic. Authors of the software are often unaware of the distinction between the two methods.
- The conflicting motivations also make it difficult to evaluate uncertainty visualisations
- In graphics research, user studies are king and allow us to understand which graphics convey information most effectively
- Uncertainty visualisation presents a unique issue. Most research tries to evaluate users ability to effectively extract information from a plot to establish a hierarchy of visualisation techniques
- Uncertainty visualisation requires signal amplification when uncertainty is low, and obfuscation when uncertainty is high. This means that the best visualisation technique is not one that allows for the most accurate statistic extraction.
- Some papers have specifically tried to evaluate uncertainty, but without a clear definition of uncertainty, this have proven to be difficult.
- This thesis attempts to close these gaps by providing a comprehensive analysis of uncertainty visualisation, including their definition, practical application, and evaluation.
  - First we comb the literature and discuss the motivations behind uncertainty visualisation, and identify motivations for the field.
  - Next, we formalise uncertainty visualisation, by identifying the two fields, discussing their difference and which is more appropriate to be considered “uncertainty visualisation”
  - Then, we formalise the second camp and coin the term “signal supression” to describe visualisation approaches that fall within it.
  - Then we discuss uncertainty’s place in the grammar of graphics, and describe what it means for a visualisation to be an “uncertainty visualisation” within this framework.
  - Then we introduce software, ggdibbler, which is the practical application of the theoretical inclusion of uncertainty in the grammar of graphics
  - Finally, we formalise an approach for evaluating the effectivness of uncertainty visualisation, and use this tool to evaluate existing approaches to uncertainty visualisation.

1.1 Thesis Outline

The thesis is structured as follows.

Chapter 2 provides a comprehensive review of the uncertainty visualsiation literature. Existing reviews in the field take the purpose of an uncertainty visualisation to be self evident which results in a large amount of conflicting information. The most common approach communicates uncertainty as a probability or a distribution, and focus on showing the uncertainty as an isolated variable disconnected from its context. We contrast these visualisations with approaches that view uncertainty as noise that should be incorporated to give a holistic view of our data. We coin the term "signal suppression" to describe a visualisation that is designed for preventing false conclusions, as the approach demands that the signal (i.e. the conclusions drawn from the estimates) is suppressed by the noise (i.e. the variance on those estimates). We provide motivation for viewing signal supression as the most worthwhile goal in uncertainty visualisation, and argue that visualisations that display uncertainty as an isolated variable should not be considered uncertainty visualisations at all. We further discuss difficulties in creating and evaluating the effectiveness of these plots, which further motivates the following chapters of this thesis.

Chapter 3 introduces a new R package, ggdibbler. In Chapter 2 we discussed the run on effects of the ambiguous definition of uncertainty. These problems included difficulty incorperating uncertainty into the grammar of graphics framework, which leads to software that does not always behave in ways that users would expect. Chapter 3 closes this gap by discussing the theoretical implications of including uncertainty in the grammar of graphics, and introducing ggdibbler, a ggplot extension that allows us to visualise uncertainty holistically and prevent false signals. We illustrate how ggdibbler can be seamlessly integrated into existing visualisation workflows and demonstrate its usefulness by looking at examples that use ggdibbler to incorporate uncertainty into a choropleth map.

Chapter 4 presents a user study, motivated by problems faced by the Australian Energy Market Operator (AEMO), that evaluates the effectiveness of different uncertainty visualisations. Chapter 2 highlighted the difficulty in evaluating uncertainty visualisations, as the uncertainty must be evaluated as noise rather than signal, and provides some suggestions for experimental methods. This section applies the suggested experimental methods and identifies which graphics, implemented by the ggdibbler software introduced in Chapter 3, are effective tools for signal suppression. The experiment in this section is motivated by a case study specific to problems faced by AEMO. The transition to renewable energy results in a large amount of uncertainty in Australia's energy demands, which also demands better understanding of the uncertainty if we are to make the transition without issue.

Chapter 2 is an invited contribution to Volume 14 of *Annual Reviews of Statistics and it's Applications*. The ggdibbler software introduced in Chapter 3 has been submitted to CRAN, and the associated paper will be submitted to the Journal of Computational and Graphical Statistics upon it's completion.

Chapter 2

The Noisy Work of Uncertainty Visualisation: A Review

This chapter contains

Chapter 3

Software for Simple Uncertainty Visualisation

3.1 Introduction

- Reiterate some points from previous chapter
- More focus on grammar of graphics

3.2 Including uncertainty in The Grammar of Graphics

- Not immediately clear where uncertainty should fit in the grammar of graphics *Get discussion from old paper*

3.3 Existing tools for visualising uncertainty

- ggdist is the most obvious implementation of uncertainty in the grammar of graphics
  - takes the same approach as ggdibbler (i.e. takes a distribution as an input) *Get discussion from mid milestone report*

3.4 ???

Check Journal of Computational and Graphical Statistics for general format

3.5 Future Work

- Different types of plots (e.g. scatter plot)
- Different data types for the uncertainty. Currently only have uncertainty in estimates, but could have uncertainty in location in a map for example.
- Alternative variations, e.g. bivariate map.
  - Could implement these methods across different plot types. e.g. what does the space transformation in the bivariate map look like when the x and y component are both spatial instead of coloured.
- Entirely new approaches
- Want to add dibble object so that it is automatic.
- Should include density objects as a “stat_identity” option for the sake of completeness.

Chapter 4

Evaluation Methods for Signal

4.1 Evaluation methods for uncertainty visualisations

Existing user evaluations don't work good.

4.2 Assessing the impact of noise on a visualisation

How do you measure the effect of noise in a visualisation?

4.3 User study: what makes a good choropleth map?

Results

Chapter 5

Conclusion

This chapter contains

5.1 Contributions

The primary contributions of this research are

5.2 Future work

There are several avenues one could take to improve upon this work.

Bibliography

Ushey, K (2022). *renv: Project Environments*. R package version 0.16.0. https://CRAN.R-project.org/package=renv.